

Perfect competition, Cournot and Stackelberg

Different market structures

Consider a market with inverse demand

$$P = a - bQ \quad Q = q_1 + q_2$$

and two firms with cost functions

$$C_1(q_1) = c_1q_1 \quad C_2(q_2) = c_2q_2$$

where

$$c_1 < c_2 < a$$

Analyze the following market structures:

1. Perfect competition
2. Cournot competition with two firms
3. Stackelberg competition with two firms, where firm 1 is the leader

For each case, calculate the market price, the quantity produced by each firm, total quantity, each firm's profit, total profits, consumer surplus, and total surplus.

1. Perfect competition

Assume firms are price takers. Find the competitive equilibrium price and quantities. Then calculate profits, consumer surplus, and total surplus.

2. Cournot competition

Assume both firms choose quantities simultaneously.

- (a) Write each firm's profit function
- (b) Derive the best-response functions
- (c) Graph the best responses in the (q_1, q_2) plane
- (d) Find the Cournot-Nash equilibrium quantities
- (e) Calculate the equilibrium price, profits, consumer surplus, and total surplus
- (f) Briefly interpret the result

3. Stackelberg competition

Assume firm 1 chooses its quantity first and firm 2 observes q_1 before choosing q_2 .

- (a) Write firm 2's best-response function
- (b) Solve firm 1's problem by backward induction
- (c) Graph the Stackelberg equilibrium in the (q_1, q_2) plane
- (d) Find the Stackelberg equilibrium quantities
- (e) Calculate the equilibrium price, profits, consumer surplus, and total surplus
- (f) Briefly interpret the result and compare it with the Cournot outcome

Solution

1. Perfect competition

Under perfect competition, firms take the market price p as given. Firm i 's profit is

$$\pi_i(p, q_i) = pq_i - c_i q_i = (p - c_i)q_i$$

Since firm 1 has the lower marginal cost, the competitive price is

$$p^{PC} = c_1$$

At this price, firm 2 does not produce because

$$p^{PC} = c_1 < c_2$$

Therefore,

$$q_2^{PC} = 0$$

Using demand,

$$c_1 = a - bQ \iff Q^{PC} = \frac{a - c_1}{b}$$

Since firm 2 produces zero, firm 1 supplies the whole market:

$$q_1^{PC} = \frac{a - c_1}{b} \quad q_2^{PC} = 0 \quad Q^{PC} = \frac{a - c_1}{b}$$

Profits are

$$\pi_1^{PC} = (p^{PC} - c_1)q_1^{PC} = 0 \quad \pi_2^{PC} = (p^{PC} - c_2)q_2^{PC} = 0$$

so total profits are

$$\Pi^{PC} = 0$$

Consumer surplus is

$$CS^{PC} = \frac{1}{2}(a - p^{PC})Q^{PC} = \frac{1}{2}(a - c_1)\frac{a - c_1}{b} = \frac{(a - c_1)^2}{2b}$$

Total surplus is

$$TS^{PC} = CS^{PC} + \Pi^{PC} = \frac{(a - c_1)^2}{2b}$$

Perfect-competition outcome:

$$p^{PC} = c_1 \quad q_1^{PC} = \frac{a - c_1}{b} \quad q_2^{PC} = 0 \quad Q^{PC} = \frac{a - c_1}{b}$$

$$\pi_1^{PC} = 0 \quad \pi_2^{PC} = 0 \quad \Pi^{PC} = 0$$

$$CS^{PC} = \frac{(a - c_1)^2}{2b} \quad TS^{PC} = \frac{(a - c_1)^2}{2b}$$

2. Cournot competition

Each firm chooses quantity taking the other firm's quantity as given.

Firm 1's profit is

$$\pi_1(q_1, q_2) = (a - b(q_1 + q_2))q_1 - c_1 q_1 = (a - c_1)q_1 - bq_1^2 - bq_1 q_2$$

Firm 2's profit is

$$\pi_2(q_1, q_2) = (a - b(q_1 + q_2))q_2 - c_2q_2 = (a - c_2)q_2 - bq_2^2 - bq_1q_2$$

The first-order conditions are

$$\frac{\partial \pi_1}{\partial q_1} = a - c_1 - 2bq_1 - bq_2 = 0 \quad \frac{\partial \pi_2}{\partial q_2} = a - c_2 - bq_1 - 2bq_2 = 0$$

Thus, the best-response functions are

$$BR_1(q_2) = \frac{a - c_1 - bq_2}{2b} \quad BR_2(q_1) = \frac{a - c_2 - bq_1}{2b}$$

Equivalently,

$$q_1 = \frac{a - c_1}{2b} - \frac{q_2}{2} \quad q_2 = \frac{a - c_2}{2b} - \frac{q_1}{2}$$

Solving the system,

$$q_1^C = \frac{a - 2c_1 + c_2}{3b} \quad q_2^C = \frac{a + c_1 - 2c_2}{3b}$$

The assumption $a > 3c_2 - 2c_1$ implies $a + c_1 > 2c_2$, so both quantities are positive.

Total output is

$$Q^C = q_1^C + q_2^C = \frac{2a - c_1 - c_2}{3b}$$

The equilibrium price is

$$p^C = a - bQ^C = a - b \frac{2a - c_1 - c_2}{3b} = \frac{a + c_1 + c_2}{3}$$

Using the first-order conditions, equilibrium profits can be written as

$$\pi_i^C = b(q_i^C)^2$$

Therefore,

$$\pi_1^C = \frac{(a - 2c_1 + c_2)^2}{9b} \quad \pi_2^C = \frac{(a + c_1 - 2c_2)^2}{9b}$$

Total profits are

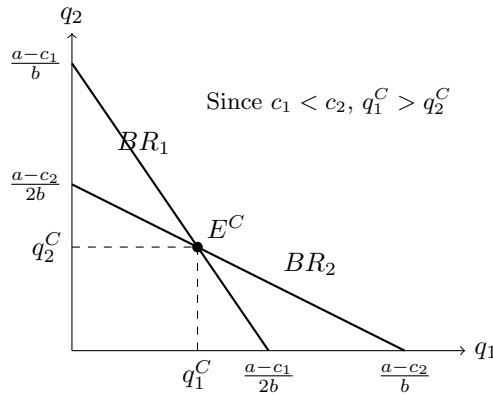
$$\Pi^C = \frac{(a - 2c_1 + c_2)^2 + (a + c_1 - 2c_2)^2}{9b}$$

Consumer surplus is

$$CS^C = \frac{1}{2}(a - p^C)Q^C = \frac{1}{2}b(Q^C)^2 = \frac{(2a - c_1 - c_2)^2}{18b}$$

Total surplus is

$$TS^C = CS^C + \Pi^C = \frac{(2a - c_1 - c_2)^2}{18b} + \frac{(a - 2c_1 + c_2)^2 + (a + c_1 - 2c_2)^2}{9b}$$



Since $c_1 < c_2$, firm 1 is more efficient. Indeed,

$$q_1^C - q_2^C = \frac{c_2 - c_1}{b} > 0$$

Cournot outcome:

$$\begin{aligned} q_1^C &= \frac{a - 2c_1 + c_2}{3b} & q_2^C &= \frac{a + c_1 - 2c_2}{3b} & Q^C &= \frac{2a - c_1 - c_2}{3b} \\ p^C &= \frac{a + c_1 + c_2}{3} \\ \pi_1^C &= \frac{(a - 2c_1 + c_2)^2}{9b} & \pi_2^C &= \frac{(a + c_1 - 2c_2)^2}{9b} & \Pi^C &= \frac{(a - 2c_1 + c_2)^2 + (a + c_1 - 2c_2)^2}{9b} \\ CS^C &= \frac{(2a - c_1 - c_2)^2}{18b} \\ TS^C &= \frac{(2a - c_1 - c_2)^2}{18b} + \frac{(a - 2c_1 + c_2)^2 + (a + c_1 - 2c_2)^2}{9b} \end{aligned}$$

Interpretation: firm 1 produces more than firm 2 because it has lower marginal cost, and the Cournot price is above the competitive price

3. Stackelberg competition

Firm 1 is the leader and firm 2 is the follower. We solve the game by backward induction.

Firm 2's profit is

$$\pi_2(q_1, q_2) = (a - b(q_1 + q_2))q_2 - c_2q_2 = (a - c_2)q_2 - bq_1q_2 - bq_2^2$$

Firm 2's first-order condition is

$$\frac{\partial \pi_2}{\partial q_2} = a - c_2 - bq_1 - 2bq_2 = 0$$

Therefore, firm 2's best-response function is

$$BR_2(q_1) = \frac{a - c_2 - bq_1}{2b}$$

Firm 1 anticipates this reaction. Substituting $BR_2(q_1)$ into its profit:

$$\pi_1(q_1, BR_2(q_1)) = \left(a - b \left(q_1 + \frac{a - c_2 - bq_1}{2b} \right) - c_1 \right) q_1$$

Simplifying,

$$\pi_1(q_1, BR_2(q_1)) = \frac{a + c_2 - 2c_1 - bq_1}{2} q_1$$

Firm 1's first-order condition is

$$\frac{d\pi_1}{dq_1} = \frac{a + c_2 - 2c_1 - 2bq_1}{2} = 0$$

Thus,

$$q_1^S = \frac{a + c_2 - 2c_1}{2b}$$

Using firm 2's reaction function,

$$q_2^S = \frac{a - c_2 - bq_1^S}{2b} = \frac{a - c_2 - \frac{a + c_2 - 2c_1}{2}}{2b} = \frac{a + 2c_1 - 3c_2}{4b}$$

The assumption $a > 3c_2 - 2c_1$ guarantees that

$$q_2^S > 0$$

Total output is

$$Q^S = q_1^S + q_2^S = \frac{3a - c_2 - 2c_1}{4b}$$

The equilibrium price is

$$p^S = a - bQ^S = a - b \frac{3a - c_2 - 2c_1}{4b} = \frac{a + c_2 + 2c_1}{4}$$

Profits are

$$\pi_1^S = (p^S - c_1)q_1^S = \frac{(a + c_2 - 2c_1)^2}{8b}$$

$$\pi_2^S = (p^S - c_2)q_2^S = \frac{(a + 2c_1 - 3c_2)^2}{16b}$$

Total profits are

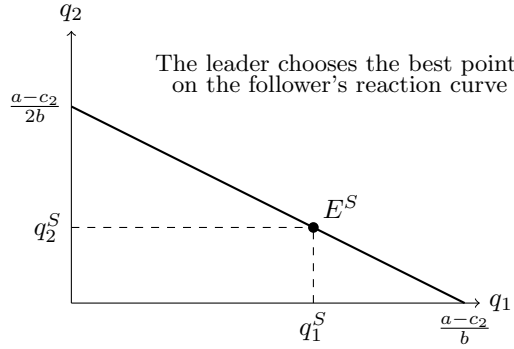
$$\Pi^S = \pi_1^S + \pi_2^S = \frac{2(a + c_2 - 2c_1)^2 + (a + 2c_1 - 3c_2)^2}{16b}$$

Consumer surplus is

$$CS^S = \frac{1}{2}(a - p^S)Q^S = \frac{1}{2}b(Q^S)^2 = \frac{(3a - c_2 - 2c_1)^2}{32b}$$

Total surplus is

$$TS^S = CS^S + \Pi^S = \frac{(3a - c_2 - 2c_1)^2}{32b} + \frac{2(a + c_2 - 2c_1)^2 + (a + 2c_1 - 3c_2)^2}{16b}$$



Comparing Stackelberg with Cournot, firm 1 produces more as leader:

$$q_1^S - q_1^C = \frac{a + c_2 - 2c_1}{6b} > 0$$

Firm 2 produces less than in Cournot:

$$q_2^S - q_2^C = -\frac{a + c_2 - 2c_1}{12b} < 0$$

Total output is higher in Stackelberg:

$$Q^S - Q^C = \frac{a + c_2 - 2c_1}{12b} > 0$$

Therefore, price is lower in Stackelberg than in Cournot:

$$p^S < p^C$$

Stackelberg outcome:

$$q_1^S = \frac{a + c_2 - 2c_1}{2b} \quad q_2^S = \frac{a + 2c_1 - 3c_2}{4b} \quad Q^S = \frac{3a - c_2 - 2c_1}{4b}$$

$$p^S = \frac{a + c_2 + 2c_1}{4}$$

$$\pi_1^S = \frac{(a + c_2 - 2c_1)^2}{8b} \quad \pi_2^S = \frac{(a + 2c_1 - 3c_2)^2}{16b}$$

$$\Pi^S = \frac{2(a + c_2 - 2c_1)^2 + (a + 2c_1 - 3c_2)^2}{16b}$$

$$CS^S = \frac{(3a - c_2 - 2c_1)^2}{32b}$$

$$TS^S = \frac{(3a - c_2 - 2c_1)^2}{32b} + \frac{2(a + c_2 - 2c_1)^2 + (a + 2c_1 - 3c_2)^2}{16b}$$

Interpretation: Stackelberg gives firm 1 a first-mover advantage. Compared with Cournot, firm 1 produces more, firm 2 produces less, total output is higher, and the market price is lower

Final comparison across market structures

Under the assumption $a > 3c_2 - 2c_1$, so that both firms produce positive quantities in Cournot and Stackelberg, the outcomes are summarized as follows:

Variable	Perfect competition	Cournot	Stackelberg
p	c_1	$\frac{a + c_1 + c_2}{3}$	$\frac{a + c_2 + 2c_1}{4}$
q_1	$\frac{a - c_1}{b}$	$\frac{a - 2c_1 + c_2}{3b}$	$\frac{a + c_2 - 2c_1}{2b}$
q_2	0	$\frac{a + c_1 - 2c_2}{3b}$	$\frac{a + 2c_1 - 3c_2}{4b}$
Q	$\frac{a - c_1}{b}$	$\frac{2a - c_1 - c_2}{3b}$	$\frac{3a - c_2 - 2c_1}{4b}$

Variable	Perfect competition	Cournot	Stackelberg
π_1	0	$\frac{(a - 2c_1 + c_2)^2}{9b}$	$\frac{(a + c_2 - 2c_1)^2}{8b}$
π_2	0	$\frac{(a + c_1 - 2c_2)^2}{9b}$	$\frac{(a + 2c_1 - 3c_2)^2}{16b}$
Π	0	$\frac{(a - 2c_1 + c_2)^2 + (a + c_1 - 2c_2)^2}{9b}$	$\frac{2(a + c_2 - 2c_1)^2 + (a + 2c_1 - 3c_2)^2}{16b}$
CS	$\frac{(a - c_1)^2}{2b}$	$\frac{(2a - c_1 - c_2)^2}{18b}$	$\frac{(3a - c_2 - 2c_1)^2}{32b}$
TS	$\frac{(a - c_1)^2}{2b}$	$\frac{(2a - c_1 - c_2)^2}{18b} + \frac{(a - 2c_1 + c_2)^2 + (a + c_1 - 2c_2)^2}{9b}$	$\frac{(3a - c_2 - 2c_1)^2}{32b} + \frac{2(a + c_2 - 2c_1)^2 + (a + 2c_1 - 3c_2)^2}{16b}$

Main comparison: perfect competition yields the lowest price and the highest total output, Stackelberg lies between perfect competition and Cournot, and Cournot yields the highest price and the lowest total output among the three cases

$$p^{PC} < p^S < p^C \quad Q^{PC} > Q^S > Q^C$$